Searching for CNN Architectures for Remote Sensing Scene Classification

Clifford Broni-Bediako, *Student Member, IEEE*, Yuki Murata, Luiz H. B. Mormille, and Masayasu Atsumi

***Abstract*— Convolutional neural network (CNN) models for remote sensing (RS) scene classification are largely built on pretrained networks that are trained on the general-purpose ImageNet dataset in computer vision. The pretrained networks can easily be adapted for transfer learning in RS scene classification. However, the accuracy of transfer learning may decline as RS images are considerably different from other images. Thus, the pretrained CNN model learned on ImageNet may not be sufficient for the accurate classification of RS image scenes. Furthermore, most of the pretrained models have large memory footprints, which place a further burden on computational requirements. In this work, we explore SLGE-based random search with early stopping in the search for CNN architectures for both single-label and multilabel RS scene classification tasks. In SLGE, the architecture search space is capable of representing multipath Inception-like modular cells with skip-connections similar to human-expert designs. The experimental results on four RS scene classification benchmarks show that the automatically discovered networks demonstrate the promising capability in classifying multispectral satellite image scenes compared with fine-tuned pretrained CNN models. Using fewer parameters with 0.56B FLOPS, our best network achieves a classification accuracy rate of 96.56% and 96.10% on NWPU-RESISC45 single-label and AID single-label RGB aerial image datasets, respectively, and the classification accuracy rate of 99.76% and 93.89% on EuroSAT single-label and BigEarthNet multilabel multispectral satellite image datasets, respectively. The results position our approach among the best of the state of the art.**

***Index Terms*— Convolutional neural networks (CNNs), neural architecture search (NAS), random search, remote sensing (RS) scene classification.**

I. Introduction

REMOTE sensing (RS) scene classification has many important applications, including urban planning, land resource management, geospatial object detection, and environmental monitoring [1]. Vast volumes of very-high-resolution RS images have become easily available due to progress in satellite technology [2]. For over 60 years, researchers have been applying learning algorithms in resolving different kinds of RS image classification [3]. Interpretation of RS imagery is a complex and tedious task. Until relatively recent years, even a simple task, such as identifying building footprints, requires laborious subspecialities. Deep learning [4] has opened up an entirely novel frontier of learning algorithms, including convolutional neural network (CNN) techniques that have been adopted in the RS research community. CNNs are capable of learning hierarchical and insightful features from large volumes of image data automatically.

Since 2014, there has been a significant progress in RS scene classification using CNN models [5]–[7]. Notwithstanding the remarkable success that CNN-based methods have achieved in RS scene classification, the design of the network architectures, which is crucial for the performance of the CNN models, still presents a huge challenge. The state-of-the-art CNN architecture design requires expertise in both deep learning methodology and domain knowledge [8]. Apart from the fact that handcrafted architecture design is time-consuming and error-prone, experts with considerable knowledge in both deep learning methodology and the investigated domain are often not available in practice.

To this end, most CNN-based RS scene classification studies employed networks [9]–[15] that are pretrained on ImageNet [16], a general-purpose image dataset in computer vision. Since both tasks involve image data, the pretrained networks on ImageNet can readily be used for transfer learning in RS scene classification. However, RS images are considerably different with respect to spectral bands, spatial resolution, and radiometric resolution [17]. Yosinski *et al.* [18] have highlighted that the effectiveness of transfer learning declines as the differences in the primary and target tasks widens. Thus, the pretrained networks learned on ImageNet may not be sufficient for RS scene classification. Moreover, most of the pretrained networks are bulky with a very high memory footprint [19], which is prohibitively expensive for resource-constrained devices, such as drones in RS applications [20]. Hence, the need to design specific CNN architectures that are suitable for RS scene classification applications is very important.

Recent dramatic shift from expert-handcrafted architecture engineering to automated task-specific architecture learning based on neural architecture search (NAS) [21] has already yielded the state-of-the-art results in computer vision and natural language processing (NLP) tasks [22]–[24]. For the first time in RS image classification task, Chen *et al.* [25] have demonstrated the potential of NAS method for searching CNN architecture for hyperspectral image classification.

In this article, we present the first effort toward applying symbolic linear generative encoding (SLGE)-based automated architecture search [26] with random search strategy

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to both single-label and multilabel RS scene classification tasks. Random search provides a strong competitive baseline for hyperparameter search problem [27]. Leveraging random search with SLGE as a specialized hyperparameter search problem, we search for the best CNN architecture on the NWPU-RESISC45 single-label dataset [1] and then apply this architecture to the AID single-label dataset [28], the EuroSAT single-label dataset [29], and the BigEarthNet large-scale multilabel dataset [30]. The key contributions of this article are summarized as follows.

1) Using SLGE-based architecture representation, we explore random search with early stopping strategy for the first time in the search for CNN architectures for RS scene classification tasks.
2) We evaluate the architecture transferability of automatically generated networks on a single-label RGB aerial image dataset to another single-label RGB aerial image dataset and to multispectral satellite image datasets of single-label and multilabel scene classification tasks.
3) The best networks discovered on NWPU-RESISC45 single-label RGB aerial image dataset compete with the state-of-the-art models on the AID single-label RGB aerial image dataset and improve the performance on EuroSAT single-label and BigEarthNet multilabel multispectral satellite image datasets.
4) The experimental results show that the proposed architecture search method is a competitive method for RS scene classification tasks, particularly for multispectral image scene classification. Our best network achieves 96.56% classification accuracy on NWPU-RESISC45, and when evaluated on AID, EuroSAT, and BigEarthNet, it achieves classification accuracy of 96.10%, 93.89%, and 99.76%, respectively, using fewer parameters with 0.56B FLOPS.

The remainder of this article is organized as follows. Section II reviews related works, while Section III explains the proposed approach for RS scene classification. The experimental results are reported in Section IV and discussed in Section V. Section VI provides some concluding remarks.

II. Related Work

A. NAS With Random Search

Automated machine learning (AutoML) methods [31], particularly NAS, enable domain scientists to automatically learn task-specific CNN architectures to solve their domain problems with little or no expertise in deep learning. Various search strategies have been adopted in NAS to learn CNN architectures for computer vision and NLP problems. This includes reinforcement learning [32], [33], evolutionary algorithms [8], [23], gradient-based methods [24], [34], Bayesian optimization [35], and random search [36], [37]. Elsken *et al.* [21] and Wistuba *et al.* [38] have provided comprehensive survey on recent NAS methods.

Random search strategy is the simplest hyperparameter optimization method [39]. Bergstra and Bengio [27] have shown that random search is able to find models that are good or better within a small fraction of the computation time compared with grid search. Random search can find better models by effectively searching a larger, less promising search space [27]. Many open-source hyperparameter optimization tools implement random search, including Vizier [40] and AHSA [41] used in Chen *et al.* [36] and Li and Talwalkar *et al.* [42], respectively, to find CNN architecture via random search.

Certainly, sophisticated NAS search algorithms, such as reinforcement learning [22], evolutionary algorithm [23], and gradient-based methods [24], have achieved remarkable results. However, as highlighted in [43], a simple random search is competitive with the state-of-the-art reinforcement learning [22], [44] and other NAS methods [23], [24], [45]. The efficiency of random search has further been explored using classical random graph models to generate CNN architectures [37]. The extensive experiments performed by [42] and [43] shown that random search is sufficiently efficient for searching CNN architectures for several dataset in computer vision and NLP using less computation resources and easy to implement.

B. Scene Classification With CNNs

Most CNN-based RS scene classification studies employed pretrained networks, such as AlexNet [9], CaffeNet [10], VGGNet [12], GoogLeNet (Inception) [13], and RestNet [15]. The pretrained networks are used as feature extractors or fine-tuned using transfer learning approach. GLANet [46], DLGFF [47], GCF-LOF CNN [48], and FDPResNet [49] are among the state-of-the-art models that have adopted pretrained CNNs networks as feature extractors for single-label scene classification.

Other works focus on fine-tuning the pretrained networks on small-scale single-label RS image datasets [50]–[52]. Zeggada *et al.* [53] and Koda *et al.* [54] applied radial basis functions and support vector machines, respectively, on CNN features as multilabel RS scene classifiers. Integration of attention-based recurrent neural networks into pretrained feature extraction networks has also been introduced in multilabel RS scene classification [55]–[57].

To depart from pretrained networks, Liu and Huang [58] proposed a weakly supervised triplet network using weakly single-label data to train CNN from the scratch. Stivaktakis *et al.* [59] introduced a data augmentation strategy to train shallow CNNs on a multilabel variation of the UC Merced dataset. In K-Branch CNN [60], spatial resolution-specific CNN branches with multiattention strategy is proposed for BigEarthNet [30] multilabel classification benchmark. Chen *et al.*, for the first time in RS, used the architecture search space that was considered by DARTS [24] with the gradient-based NAS strategy to find CNN architectures for hyperspectral image classification [25] and image scene classification [61] tasks.

Motivated by these various architecture designs for RS scene classification and the efficiency of random search, we adopt the architecture representation introduced by SLGE [26] and random search strategy as an NAS method

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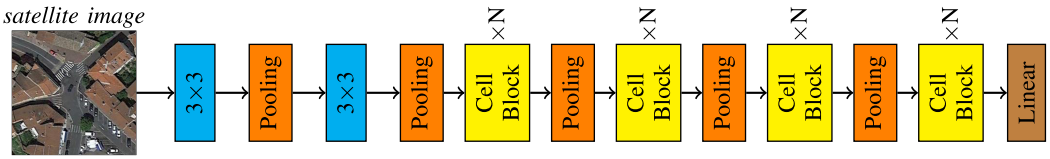


Fig. 1. Entire CNN architecture is constructed by stacking an autogenerated cell (see Fig. 3) as *cell blocks* marked in yellow. The architecture begins with two 3 × 3 convolution layers followed by several autogenerated cell blocks repeated N times with average pooling layers inserted in between them to downsample the spatial resolutions.

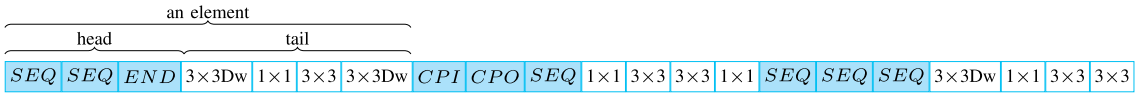


Fig. 2. Example of SLGE linear structure. SEQ, CPO, CPI, and END are transformation functions, and 1 × 1, 3 × 3, and 3 × 3Dw (depthwise) are convolution operations. (We reference [26] for more details.)

to search for CNN architectures for both single-label and multilabel RS scene classification tasks.

III. Methodology

According to our SLGE [26], the problem of searching for CNN architectures for RS scene classification is formulated as follows. Given the problem space $\psi = \{\mathcal{A}, \mathcal{S}, \mathcal{P}, t\mathcal{D}, v\mathcal{D}\}$, where $\mathcal{A}$ is the architecture search space, $\mathcal{S}$ represents the search strategy, $\mathcal{P}$ denotes the performance measure, and $t\mathcal{D}$ and $v\mathcal{D}$ are the training and validation datasets, respectively, the objective is to find a small CNN architecture $a^* \in \mathcal{A}$ via the random search strategy $\mathcal{S}$, which, after training it on the dataset $t\mathcal{D}$, maximizes the performance measure of classification accuracy $\mathcal{P}$ on the validation dataset $v\mathcal{D}$. Small architecture here means a CNN model, which has the number of parameters θ less than or equal to the target size. Mathematically, the objective function $\mathcal{F}$ can be formulated as

$$\mathcal{F}(\psi) = \max_{\theta,\ a} \mathcal{P}(\mathcal{L}(a(\theta), t\mathcal{D} \mid \mathcal{S}, a \in \mathcal{A}), v\mathcal{D}) \quad \text{s.t. the model size of } (a(\theta)) \leq \mathcal{T}_{\text{params}} \tag{1}$$

where $\mathcal{L}$ represents the training of the model parameters θ with the loss function and $\mathcal{T}_{\text{params}}$ denotes the target number of parameters.

Following previous studies on NAS [23], [33], the SLGE-based random search searches for the topological structure of cells that are repeatedly stacked to construct an entire CNN architecture (see Fig. 1). The cell is a directed acyclic graph (DAG) generated via the search strategy $\mathcal{S}$ from the architecture search space $\mathcal{A}$. Each node of the cell represents a convolutional layer, whereas edges are the representation of latent information flow direction in the CNN architecture. The architecture search space and search strategy are the two key components of the NAS method [21]. To reduce the computational cost, the proxy task is normally employed on the search space to search for good architectures [22], [62]. We now describe the search space, search strategy, and proxy task that we adopt in this work.

A. Search Space

In this work, we use the search space that was introduced by SLGE [26]. The search space represents individual architectures as linear string structures. A mapping function is used to translate the linear structure into a cell (DAG), which is typically stacked repeatedly to build a final CNN architecture in Fig. 1. The search space can produce multipath Inception-like [13] cells with skip-connections [15] similar to human-expert designs. Fig. 2 is an example of a linear string structure in the search space, which is converted into a cell shown in Fig. 3. We refer the interested readers to [26] for more details.

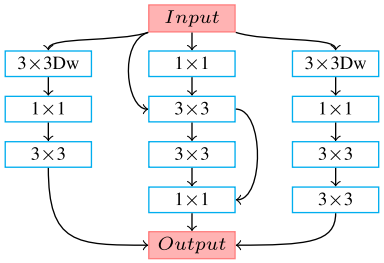


Fig. 3. Example of SLGE multipath Inception-like cell (DAG) with skip-connections. A cell representation of the linear string is shown in Fig. 2. The convolution operations without successor are depthwise concatenated to produce an output, and if an operation has more than one predecessor, the feature maps of the predecessors are added together. This is a common approach in cell-based search [33].

In SLGE, three types of convolution operations conv3 × 3, conv1 × 1, and conv3 × 3 depthwise are used as cell nodes. The convolution operation nodes without successor are depthwise concatenated to produce the *output* in a cell (see Fig. 3), and if a node has more than one predecessor, the outputs of the predecessors are added together. This is a common approach in the NAS literature [22], [33]. We adopt four different search spaces of SLGE linear string structures (see Table I). This defines the overall search space with nodes between 2 and 16 in a cell and contains approximately 2.4×10^6 possible cells in a 16-node case.

B. Search Strategy

We adopt random search with early stopping strategy [63] as the search algorithm. The random search basically employs the strategy of sampling a cell uniformly at random from the search space. Then, the sampled cell is stacked repeatedly to build candidate architecture in the proxy task

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TABLE I
FOUR DIFFERENT SEARCH SPACES THAT WE ADOPT FROM SLGE (WE REFERENCE [26] FOR MORE DETAILS)

	SLGE Linear Structure		
Search Space	Element	Head	Tail
A	3	2	3
B	3	3	4
C	4	2	3
D	4	3	4

(see Section III-C) and trained using the hyperparameters presented in Section IV-B. Early stopping [63] is applied for training the candidate architectures (proxy networks) not to convergence. To make the architecture search result robust, we repeat the search several times with different random seeds.

C. Proxy Task

Searching for large-scale CNN architectures directly on large datasets, such as NWPU-RESISC45 and BigEarthNet, is computationally expensive. It takes several days for each architecture to converge. To reduce the number of computation resources required during the architecture search, we employed a small-scale task as a proxy task for quick-search evaluation. This may provide a predictive signal about training the large-scale architecture settings.

Following previous studies in computer vision [22], [23], [33], we designed a proxy task by employing a small-scale network, which consists of few repeated cell blocks in Fig. 1 and resizes the RS images into 32×32 low resolution to perform the architecture search over small networks. The proxy task network begins with only one 3×3 convolution stem with output channel size C, followed by three cell blocks denoted as $B = [b_1, b_2, b_3]$, and then a classifier. Each block i in B consists of the repeated cell of b_i times with the average pooling layer inserted between them to downsample the feature dimension. Average pooling is used to reduce the feature maps resolutions of each cell block by half, and then, the channel C is doubled.

We implement the proxy task with a 25 200-sample set of the NWPU-RESISC45 dataset. The sample set is split into 80% training and 20% validation subsets with normalization. To speed up the search process, proxy task networks with $C = 32$ and $B = [2, 2, 2]$, which have relatively small model memory footprint (less than 2M parameters), are used. Each generated proxy network is trained on the training subset without any data augmentation for 100 epochs with a batch size of 256 and evaluated on the validation subset to determine its classification accuracy. Then, the top proxy networks are retrained from scratch to full convergence in the large-scale architecture settings shown in Fig. 1 and present as the best CNN architectures. We implement the architectures in the PyTorch [64] deep learning framework.

It is worth mentioning that using a proxy task in the NAS process may generate an inaccurate ranking of the architectures since the networks with fewer parameters may converge faster and produce better results for a few epochs than cumbersome ones [43]. Although the networks with good performance may be abandoned, we adopt proxy tasks to show that we can find good architectures even with limited computation resources.

IV. EXPERIMENTS

In this section, we present the experiments with the proposed method described in Section III. All the architecture searches are performed on the NWPU-RESISC45 single-label classification task. A total of 400 CNN architectures are searched on an 11-GB GPU GeForce GTX1080 Ti machine using five different random seeds for 9.6 GPU days. The best architectures discovered on NWPU-RESISC45 are evaluated on AID single-label, EuroSAT single-label, and BigEarthNet multilabel scene classification tasks to verify their effectiveness. We compared the results on NWPU-RESISC45, AID, and EuroSAT with the state-of-the-art transfer learning models and an NAS-based baseline model, whereas the results on BigEarthNet are compared with the state-of-the-art CNN models trained from scratch and the baseline. The codes are available at https://github.com/cliffbb/geppy_nn.

A. Datasets

The NWPU-RESISC45[1] is a single-label dataset created by the Northwestern Polytechnic University [1]. The dataset consists of 31 500 satellite images categorized into 45 classes. Each class includes 700 images of size 256×256 in RGB space with spatial resolution varying from 30 to 0.2 m. The images were acquired from Google Earth imagery of more than 100 countries and regions.

AID[2] is made up of satellite images acquired from Google Earth imagery of different countries and regions [28]. The dataset consists of 10 000 single-label images categorized into 30 classes. The number of samples per class varies from 220 up to 420. Each image is 600×600 pixels in RGB space with spatial resolution varying from about 8 to about 0.5 m.

EuroSAT[3] is a single-label multispectral image dataset acquired from Sentinel-2 satellite images of cities in the 34 European countries [29]. The dataset consists of 10 different classes with 2000–3000 images per class. In total, the dataset has 27 000 images with 13 different spectral bands of size 64×64 pixels. We use the images of the RGB and RGB-NIR (RGB and near-infrared) 10-m bands.

BigEarthNet[4] is a multilabel dataset created from Sentinel-2 multispectral satellite images scattered over different European countries [30]. The dataset consists of 590 326 images with 12 different spectral bands. Each image is associated with one or more class labels extracted from the CORINE land cover (CLC) map of 2018. Originally, BigEarthNet has 43 classes; then, the authors realigned it in a new nomenclature of 19 classes based on CLC Level-3 nomenclature. We use the new nomenclature classes, which

[1] https://1drv.ms/u/s!AmgKYzARBl5ca3HNaHIlzp_IXjs
[2] https://captain-whu.github.io/AID/
[3] https://github.com/phelber/EuroSAT
[4] http://bigearth.net/

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is called BigEarthNet19 [65], and the RGB and RGB-NIR 10-m bands images of 120 × 120 pixels.

NWPU-RESISC45, AID, and EuroSAT are challenging RS scene classification benchmarks; however, BigEarthNet is the more complex. We randomly split NWPU-RESISC45, AID, and EuroSAT into 80/20 ratios for training and testing, respectively, and 20% of the training set is used as the validation set. The split is applied classwise. BigEarthNet is split as 269 695 for training, 123 723 for validation, and 125 866 for testing as provided by the authors. The testing sets are only used for the final performance evaluation of a model.

B. Training Details

During the architecture search, the candidate architectures are trained and evaluated on NWPU-RESISC45 based on the proxy task described in Section III-C. After the search, the cell of the best found architecture is repeatedly stacked as a four-cell block $B = [b_1, b_2, b_3, b_4]$ with initial channels $C = 16$, and average pooling layer is inserted between them to build the large-scale CNN architecture in Fig. 1, subject to model size constraint $\mathcal{T}_{\text{params}} \leq 5.5\text{M}$. We resized the images in NWPU-RESISC45 and AID to 224 × 224 but maintained the 64 × 64 and 120 × 120 size of EuroSAT and BigEarthNet, respectively, and augmented the training subset as in [15]. The large-scale architecture is trained from scratch on NWPU-RESISC45, AID, and EuroSAT for 600 epochs and on BigEarthNet for 200 epochs. We use a batch size of 128 for all the datasets.

We use fastai[5] [66], a PyTorch-based library, for the training of the architectures. All the architectures, during search and after search, are trained using one-cycle policy in [67] with Adam optimizer [68]. The learning rate is set to go from 0.0004 to 0.01 linearly, while the momentum goes from 0.95 to 0.85 linearly in phase one of the one-cycle policy. Then, in phase two, the learning rates follow cosine annealing from 0.01 to 0, as the momentum goes from 0.85 to 0.95 with the same annealing, and the weight decay is set to 0.0001. These are default values in one-cycle policy [67].

C. Baseline NAS-Based Network

We select EfficientNets [69], a gradient-based NAS model pretrained on ImageNet, as a baseline for the study. For a fair comparison, we adopt EfficientNet-B0[6] model that has 5.3M parameters [69] comparable to those of our best networks. The experimental results on the four datasets (NWPU-RESISC45, AID, EuroSAT, and BigEarthNet) are compared with EfficientNet-B0 trained from scratch and with fine-tuning from the parameters learned on ImageNet. The linear classifier of EfficientNet-B0 was simply replaced with a linear classifier, which corresponds to the number of classes in the considered datasets. EfficientNet-B0 was fully trained from scratch with 80/20 training–testing ratio using the hyperparameters described in Section IV-B. On fine-tuning, the replaced linear classifier was trained for 100 epochs with the same training settings in Section IV-B. Then, we halved the learning rate and trained the whole model for 150 epochs. The baseline was fine-tuned with both 20/80 and 80/20 training–testing ratios on NWPU-RESISC45 and 50/50 and 80/20 training–testing ratios on AID. On EuroSAT and BigEarthNet, it was fine-tuned with only 80/20 training–testing ratios.

[5]https://fastai1.fast.ai/

[6]https://github.com/lukemelas/EfficientNet-PyTorch

TABLE II

RESULTS OF TOP-20 NETWORKS FOUND VIA THE PROXY TASK AND FULLY TRAINED ON NWPU-RESISC45 IN THE LARGE-SCALE ARCHITECTURE SETTINGS (SEE FIG. 1). THE BOLDFACE IS THE BEST RESULT

Search Space	Seed	Cell Block	Accuracy (%)	Model Size
A	320	[4, 5, 5, 3]	95.37±0.09	5.3M
	321	[4, 4, 4, 3]	95.12±0.19	5.5M
	322	[3, 2, 1, 2]	**96.12±0.18**	5.5M
	323	[3, 2, 1, 2]	95.45±0.31	5.4M
	324	[3, 3, 1, 2]	95.74±0.23	5.5M
B	330	[2, 2, 1, 1]	96.02±0.14	5.0M
	331	[2, 3, 3, 1]	**96.36±0.11**	5.4M
	332	[2, 1, 2, 1]	96.13±0.14	5.4M
	333	[3, 4, 4, 1]	95.91±0.21	5.3M
	334	[4, 4, 4, 3]	95.72±0.20	5.0M
C	420	[4, 3, 4, 3]	96.14±0.15	5.5M
	421	[2, 2, 1,21]	95.73±0.18	5.5M
	422	[2, 3, 2, 1]	**96.56±0.13**	5.1M
	423	[2, 3, 3, 2]	95.37±0.86	5.2M
	424	[3, 4, 4, 3]	95.81±0.10	5.5M
D	430	[2, 2, 1, 1]	96.08±0.15	5.1M
	431	[3, 4, 4, 1]	**96.44±0.21**	5.5M
	432	[2, 2, 1, 1]	96.02±0385	5.0M
	433	[2, 2, 1, 1]	96.28±0.11	5.3M
	434	[2, 2, 2, 1]	96.21±0.13	5.0M

D. Searching on NWPU-RESISC45

To investigate the effectiveness of the proposed method, we conducted several experiments on four different search spaces that we adopted from SLGE (see Table I). For each search space, the experiment was performed five times with different random seeds. We randomly sampled 20 individual cells per experiment to build 20 different proxy architectures and then trained and evaluated based on the proxy task (see Section IV-B). A total of 400 individual cells were sampled to build 400 different architectures during the architecture search process. Fig. 4 shows the search results of the proxy task for each search space. The results show an average accuracy of 82.23% with a 1.72% standard deviation over the 400 proxy networks. Although the networks in the proxy task were not trained to converge, the results surpassed the feature extraction method in [1] and compete with the fine-tuning approach in [52].

The cells of top-20 networks (one from each of the five experiments per a search space) found during the architecture

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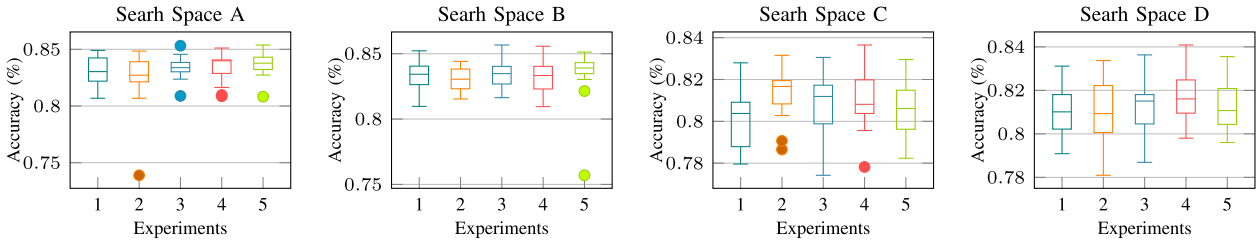


Fig. 4. Search results of the 400 proxy networks trained according to the proxy task described in Section III-C. In each search space, we performed five experiments by sampling 20 individual cells per experiment to build 20 different proxy architectures and trained each for 100 epochs. The results show the accuracy distribution of 20 networks per experiment in each search space. It took 9.6 GPU days to train and evaluate all the 400 proxy networks.

TABLE III

COMPARISONS BETWEEN OUR METHOD AND THE BASELINE NETWORK AND THE STATE-OF-THE-ART TRANSFER LEARNING (FEATURE EXTRACTION AND FINE-TUNING) METHODS WITH 20/80 TRAINING–TESTING RATIO ON THE NWPU-RESISC45 DATASET. HERE, OUR METHOD IS AUTOGENERATED ARCHITECTURES DISCOVERED ON NWPU-RESISC45. THE BASELINE RESULTS ARE BASED ON OUR TRAINING. BOLDFACE IS THE BEST RESULT

Model		Accuracy (%)	Model Size	FLOPS	Method
BoCF with VGGNet-16 [70]		84.32±0.17	138M†	15.5B†	Feature Extraction (Handcrafted)
SAL-TS-Net fusion with GoogLeNet [71]		87.01±0.19	13.6M†	2.80B†	
D-CNN with VGGNet-16 [72]		91.89±0.22	138M†	15.5B†	
DLGFF with CaffeNet [47]		96.37±0.05	60M†	0.72B†	
DDRL-AM with ResNet-18 [73]		92.46±0.09	11.5M†	1.82B†	
GLANet with VGGNet-16 [46]		93.45±0.17	138M†	15.5B†	
FDPResNet with ResNet-50 [49]		95.40±0.11	25.6M†	4.12B†	
VGGNet-16 with SVM [1]		90.36±0.18	138M†	15.5B†	Fine-Tuning (Handcrafted)
VGGNet-16 with XGBoost [52]		83.35	138M†	15.5B†	
Fusion of VGGNet-16 & CaffeNet [74]		95.36±0.22	198M†	16.2B†	
Fusion of ResNet-50 & VGGNet-16 [75]		94.03	163.6M†	19.6B†	
EfficientNet-B0 [69] (baseline)		91.98*	5.3M	0.39B	Fine-Tuning (Auto-Generated)
		96.43‡	5.3M	0.39B	
		95.90‡	5.3M	0.39B	Fully-Trained (Auto-Generated)
Our method (C=16)	B=[3, 2, 1, 2]	96.12±0.18‡	5.5M	0.58B	Fully-Trained (Auto-Generated)
	B=[2, 3, 3, 1]	96.36±0.11‡	5.4M	0.91B	
	B=[2, 3, 2, 1]	**96.56±0.13**‡	5.1M	0.56B	
	B=[3, 4, 4, 1]	96.44±0.21‡	5.5M	0.73B	

† Model size and FLOPS are based on the open source backbone pre-trained networks.
* This result is based on 20/80 training-testing ratio.
‡ This results is based on 80/20 training-testing ratio.

search are used to build 20 large-scale architectures (see Fig. 1) and trained from scratch using the training settings in Section IV-B. The architectures were trained and evaluated ten times to reduce the bias effect, and the results were averaged to determine the best network for RS scene classification. The best network achieved 96.56 ± 0.13% accuracy, as shown in Table II. The results show an average accuracy of 95.93% with 0.21% standard deviation over the top-20 networks.

In Table III, we compared the results with the state-of-the-art transfer learning methods and the baseline in terms of classification accuracy, model size, and FLOPS. The state-of-the-art transfer learning methods on NWPU-RESISC45 in the literature are trained with a 20/80 training–testing ratio. For competitive comparison, we compared our results to these methods (see Table III). However, for a fair comparison, we also compared the results with the state-of-the-art handcrafted CNNs networks and gradient-based NAS networks that are trained from scratch with 80/20 training–testing ratio in [61] (see Table V).

With few parameters, our best network improved the performance of most of the popular transfer learning methods on NWPU-RESISC4. Table III shows an approximately 0.2% improvement over the state-of-the-art results reported in DLGFF [47]. By analyzing Table III, one can observe that fine-tuning the baseline with 80/20 training–testing ratio achieved a competitive result. However, training the baseline from scratch, our best network achieved an accuracy of approximately 1% higher than the baseline. Table V shows

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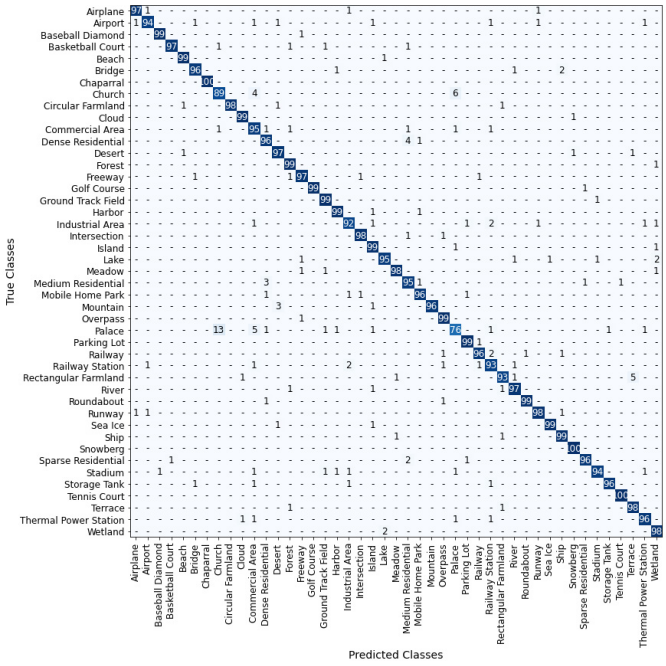


Fig. 5. Confusion matrix of our best network on NWPU-RESISC45 (see Table III). Diagonal values are correct predictions (%), nondiagonals values are incorrect predictions (%), and the dash (–) means no incorrect prediction.

that our best discovered network achieved an approximately 3% higher accuracy than gradient-based NAS model [61]. In addition, the best network improved the performance of the state-of-the-art handcrafted CNN models reported in [61] by 4% (see Table V).

To provide a further understanding of the performance of our proposed approach, we present a confusion matrix (see Fig. 5) to illustrate the correct and incorrect classification results of our best network on NWPU-RESISC4. Fig. 5 shows that our best network achieved a classification accuracy rate greater than 96% on most of the scene classes. Our best network achieved a high accuracy rate on image scenes, such as *chaparral*, *forest*, *sea ice*, and *snowberg*, which have single texture and exhibit low interclass similarity. On the other hand, scenes, such as *commercial area*, *industrial area*, and *medium residential*, that have a complex textural composition with high interclass similarity and intraclass diversity experienced low accuracy rate. The highest misclassification results occur between *palace* and *church* classes with 89% and 76% accuracy rates, respectively. Images in both classes are very similar in structural and textural characteristics (see Fig. 6).

E. Evaluating on AID

To further demonstrate the effectiveness of the proposed method, the architectures discovered on NWPU-RESISC45 were experimented on AID to evaluate the architecture transferability of the best discovered networks. AID is a single-label classification benchmark consists of RGB images, such as NWPU-RESISC45, with similar complexity and features. However, AID has only 10 000 samples and 30 scene classes,

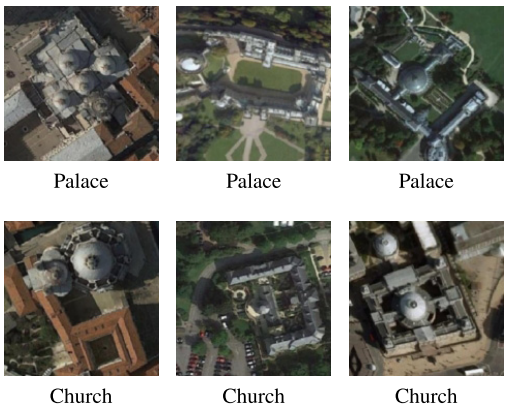


Fig. 6. Example of most confused classes, *palace* and *church*, in NWPU-RESISC45 (see Fig. 5). It shows the similarity in textural and structural features among the images.

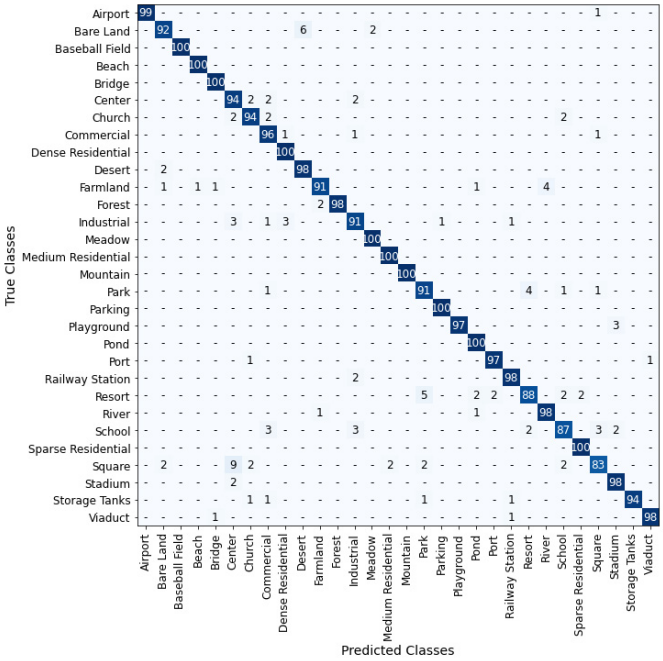


Fig. 7. Confusion matrix of our best network on AID benchmark (see Table IV). Diagonal values are correct predictions (%), nondiagonal values are incorrect predictions (%), and the dash (–) means no incorrect prediction.

which makes it a bit challenging for training the CNN model from scratch. The best four architectures discovered on NWPU-RESISC45 (see Table II) were evaluated on AID. We trained them from scratch with a linear classifier head of 30 classes. The training details remain the same as applied on NWPU-RESISC45.

The results are compared with the state-of-the-art transfer learning methods and the baseline in terms of classification accuracy, model size, and FLOPS, as shown in Table IV. In the literature, the state-of-the-art transfer learning methods on AID are trained with a 50/50 training–testing ratio. We compared the results with these methods for competitive comparison (see Table IV). Also, for a fair comparison, we compared the results with the state-of-the-art handcrafted

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TABLE IV

COMPARISONS BETWEEN OUR METHOD AND THE BASELINE NETWORK AND THE STATE-OF-THE-ART TRANSFER LEARNING (FEATURE EXTRACTION AND FINE-TUNING) METHODS WITH 50/50 TRAINING–TESTING RATIO ON THE AID DATASET. HERE, OUR METHOD IS AUTOGENERATED ARCHITECTURES DISCOVERED ON NWPU-RESISC45, WHICH ARE EVALUATED ON AID. THE BASELINE RESULTS ARE BASED ON OUR TRAINING. BOLDFACE IS THE BEST RESULT

Model		Accuracy (%)	Model Size	FLOPS	Method
AID Benchmark (VGGNet-16) [28]		89.64±0.36	138M†	15.5B†	Feature Extraction (Handcrafted)
Two-Stream Fusion (VGGNet-16) [76]		94.58±0.25	276M†	31.0B†	
SAL-TS-Net fusion with GoogLeNet [71]		95.99±0.35	13.6M†	2.80B†	
D-CNN with VGGNet-16 [72]		**96.89±0.10**	138M†	15.5B†	
GLANet with VGGNet-16 [46]		96.66±0.19	138M†	15.5B†	
Fusion of VGGNet-16 & CaffeNet (LR) [74]		94.93±0.28	198M†	16.2B†	Fine-Tuning (Handcrafted)
Fusion of VGGNet-16 & CaffeNet (SVM) [74]		95.36±0.22	198M†	16.2B†	
EfficientNet-B0 [69] (baseline)		95.54*	5.3M	0.39B	Fine-Tuning (Auto-Generated)
		96.65‡	5.3M	0.39B	
		94.35‡	5.3M	0.39B	Fully-Trained (Auto-Generated)
Our method (C=16)	B=[3, 2, 1, 2]	95.08±0.18‡	5.5M	0.58B	Fully-Trained (Auto-Generated)
	B=[2, 3, 3, 1]	95.96±0.20‡	5.4M	0.91B	
	B=[2, 3, 2, 1]	96.10±0.18‡	5.1M	0.56B	
	B=[3, 4, 4, 1]	95.49±0.16‡	5.5M	0.73B	

† Model size and FLOPS are based on the open source backbone pre-trained networks.
* This result is based on 50/50 training-testing ratio.
‡ This result is based on 80/20 training-testing ratio.

TABLE V

COMPARISONS BETWEEN OUR METHOD AND THE STATE-OF-THE-ART HANDCRAFTED CNN MODELS AND GRADIENT-BASED NAS METHOD FULLY TRAINED ON NWPU-RESISC45 (NWPU) AND AID WITH 80/20 TRAINING–TESTING RATIO. THE RESULTS OF THE STATE-OF-THE-ART HANDCRAFTED MODELS ARE REPORTED IN [61]. BOLDFACE IS THE BEST RESULT

		NWPU	AID			
Model		Accuracy (%)	Accuracy(%)	Model Size	FLOPS	Method
GoogLeNet [77]		91.61±0.65	88.15±0.77	6.8M	1.40B	Handcrafted
Inception-V3 [78]		92.58±0.26	89.03±1.27	23.2M	2.85B	
ResNet-50 [15]		92.01±0.29	88.10±0.65	25.6M	4.12B	
DenseNet [14]		92.71±0.17	87.50±0.24	13.7M	3.42B	
MobileNet-V2 [79]		92.29±0.19	89.33±0.37	6.9M	0.58B	
CNN Arch. Learning [61]		–	91.12±0.49	3.8M	–	Auto-Generated (Gradient-Based)
		93.67±0.24	–	5.2M	–	
Our method (C=16)	B=[3, 2, 1, 2]	96.12±0.18	95.08±0.18	5.5M	0.58B	Auto-Generated (Random Search)
	B=[2, 3, 3, 1]	95.96±0.20	96.36±0.11	5.4M	0.91B	
	B=[2, 3, 2, 1]	**96.56±0.13**	**96.10±0.18**	5.1M	0.56B	
	B=[3, 4, 4, 1]	96.44±0.21	95.49±0.16	5.5M	0.73B	

CNNs networks and gradient-based NAS networks that are trained from scratch with 80/20 training–testing ratio in [61] (see Table V).

Table IV shows our best network with only 5.1M parameters competed with the state-of-the-art feature extraction methods, D-CNN [72], and GLANet [46], which used VGGNet-16 as backbone pretrained network. It also improved the performance of the state-of-the-art fine-tuning methods by approximately a 1% accuracy rate. By examining Table IV, one can observe that the baseline with 80/20 training–testing ratio improved the performance of the best discovered network by approximately 0.5% when fine-tuned. However, training the baseline from scratch, the best discovered network achieved approximately 2% higher accuracy than the baseline. Table V shows the best discovered network improved the performance of the gradient-based NAS model [61] by a 5% accuracy rate. It also improved the performance of the state-of-the-art handcrafted CNN models reported in [61] by 6% (see Table V).

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TABLE VI

COMPARISONS BETWEEN OUR METHOD AND THE BASELINE NETWORK AND THE STATE-OF-THE-ART FINE-TUNED PRETRAINED BENCHMARK MODELS ON EUROSAT IN RGB SPACE AND RGB-NIR SPACE. HERE, OUR METHOD IS AUTOGENERATED ARCHITECTURES DISCOVERED ON NWPU-RESISC45, WHICH ARE EVALUATED ON THE EUROSAT DATASET. THE BOLDFACE IS THE BEST RESULT

Model		Accuracy (%) RGB	Accuracy (%) RGB-NIR	Model Size	Method
RestNet-50 [29]		98.57	–	25.6M	Fine-Tuning (Handcrafted)
GoogLeNet [29]		98.18	–	6.8M	
EfficientNet-B0 [69] (baseline)		**98.85**	98.78	5.3M	Fine-Tuning (Auto-Generated)
Our method (C=16)	B=[3, 2, 1, 2]	98.52	**99.76**	5.5M	Auto-Generated (Random Search)
	B=[2, 3, 3, 1]	98.31	99.54	5.4M	
	B=[2, 3, 2, 1]	98.67	99.63	5.1M	
	B=[3, 4, 4, 1]	98.61	99.59	5.5M	

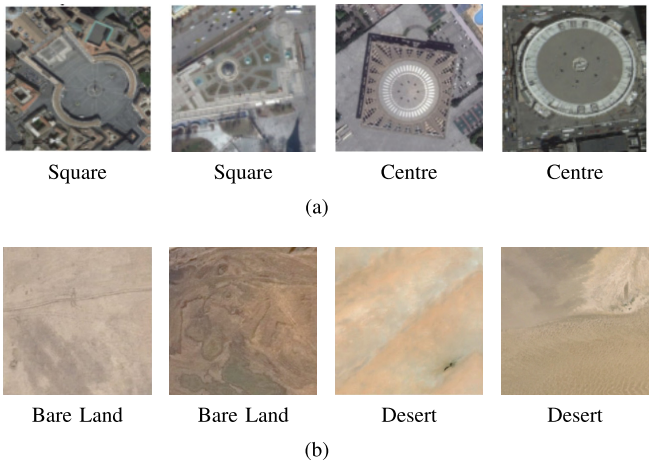


Fig. 8. Examples of confused classes in AID (see Fig. 7). This shows the textural similarity among the images: (a) *square* and *centre* classes and (b) *bare-land* and *desert* classes.

We also present a confusion matrix in Fig. 7 to illustrate the classification accuracy. Out of the 30 scene classes, 21 classes were correctly classified over 96% accuracy rate, and only three classes (*resort*, *school*, and *square*) were classified with an accuracy rate lower than 90%. The class with the lowest accuracy of 83% was the *square*, which is mostly confused with the *centre* class. Other confused scenes include *bare-land* versus *desert* and *resort* versus *park* shown in Fig. 8. These pairs of scenes are very similar in textural characteristics.

F. Evaluating on EuroSAT

We evaluated the architecture transferability of networks discovered on a single-label RGB aerial image dataset to a single-label multispectral satellite image dataset. The best four architectures discovered on NWPU-RESISC45 (see Table II) single-label RGB image dataset were evaluated on the EuroSAT multispectral image dataset. EuroSAT is a single-label classification benchmark with ten classes. Each architecture was trained from the scratch with a linear classifier head of ten classes using the training details as applied on the NWPU-RESISC45. Because the images in EuroSAT are of size 64 × 64 pixels, we did not apply the second 3 × 3 convolution and pooling layers in the large-scale architecture settings (see Fig. 1).

For comparative evaluation in Table VI, we used images in the RGB space and RGB-NIR space of the EuroSAT dataset to train our architectures discovered on NWPU-RESISC45 and also to fine-tune the baseline network. We reported in Table VI the classification accuracy to evaluate the performance of our best discovered networks and the baseline. In the RGB space, as shown in Table VI, our best discovered network achieved a classification accuracy of 98.67%, which competes with the baseline's 98.85% accuracy rate, and the 98.57% of the state-of-the-art fine-tuned pretrained benchmark models reported in [29]. However, in the RGB-NIR space, our best discovered network achieved a 99.76% accuracy rate, which is approximately 1% higher than the 98.78% of the baseline.

G. Evaluating on BigEarthNet

We also evaluated the architecture transferability of networks discovered on a single-label RGB aerial image dataset to a multilabel multispectral satellite image dataset. The best four architectures discovered on the NWPU-RESISC45 (see Table II) single-label RGB aerial image dataset were evaluated on the BigEarthNet-19 multilabel multispectral satellite image dataset. BigEarthNet-19 is a large-scale Sentinel-2 dataset with 19 classes. Since the images in BigEarthNet are of size 120 × 120 pixels, we did not apply the first pooling layer in the large-scale architecture settings (see Fig. 1). This slightly increased the size of our networks, which can be observed by comparing the model size in Tables II and VII. Each architecture was trained from scratch with a linear classifier of 19 classes. All training settings remained the same as applied on NWPU-RESISC45. In addition to classification accuracy (Acc), we reported F_1 score, recall (R), and precision (P) metrics, as described in [65].

In Table VII, we compared our results with the state-of-the-art handcrafted CNN models on the RGB images of BigEarthNet in [65]. Our best network achieved a multilabel classification accuracy of 93.49%. The best four networks that

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TABLE VII

COMPARISONS BETWEEN OUR METHOD AND THE BASELINE NETWORK AND THE STATE-OF-THE-ART HANDCRAFTED CNN MODELS ON BIGEARTHNET IN RGB SPACE AND RGB-NIR SPACE. HERE, OUR METHOD IS AUTOGENERATED ARCHITECTURES DISCOVERED ON NWPU-RESISC45, WHICH ARE EVALUATED ON THE BIGEARTHNET DATASET. THE RESULTS OF THE STATE-OF-THE-ART HANDCRAFTED MODELS ARE REPORTED IN [65]. BOLDFACE IS THE BEST RESULT

Band	Model		Acc (%)	F_1 (%)	R (%)	P (%)	Model Size	Method
RGB	VGGNet-16 [12]		–	76.01	75.85	81.05	138M	Handcrafted
	VGGNet-19 [12]		–	75.96	76.71	79.87	143M	
	ResNet-50 [15]		–	77.11	77.44	81.39	25.6M	
	ResNet-101 [15]		–	76.49	77.45	80.18	44.5M	
	ResNet-152 [15]		–	76.53	76.24	81.72	60.2M	
	K-Branch CNN [80]		–	72.73	78.96	71.61	–	
	Our method (C=16)	B=[3, 2, 1, 2]	93.34	86.36	86.76	**86.32**	5.7M	Auto-Generated (Random Search)
		B=[2, 3, 3, 1]	93.40	85.84	**86.78**	85.84	5.6M	
		B=[2, 3, 2, 1]	93.43	**86.53**	85.45	86.26	5.3M	
		B=[3, 4, 4, 1]	**93.49**	86.30	86.75	85.99	5.8M	
RGB-NIR	EfficientNet-B0 [69] (baseline)		93.03	85.29	84.69	85.68	5.3M	Fine-Tuning (Auto-Generated)
	Our method (C=16)	B=[3, 2, 1, 2]	93.66	88.09	87.60	88.58	5.7M	Auto-Generated (Random Search)
		B=[2, 3, 3, 1]	93.87	88.18	87.69	88.67	5.6M	
		B=[2, 3, 2, 1]	**93.89**	**89.21**	**88.22**	**90.23**	5.3M	
		B=[3, 4, 4, 1]	93.87	88.55	87.98	89.13	5.8M	

we trained on BigEarthNet achieved an average of 93.42% multilabel classification accuracy. Using fewer parameters, our best networks significantly improved the performance of the state-of-the-art handcrafted CNN models on BigEarthNet [65]. An improvement of approximately 9% F_1 score, 8% recall, and 4% precision was achieved over ResNet50, K-Branch CNN, and ResNet152, respectively. Moreover, for further comparative evaluation, we fine-tuned the baseline network on the RGB-NIR images of BigEarthNet. As shown in Table VII, in the RGB-NIR space, the baseline trailed behind our best network by approximately 0.9% in multilabel classification accuracy, 3.9% in F_1 score, 3.5% in recall metric, and 4.5% in precision metric. An example of the BigEarthNet RGB images with the true multilabels and the predicted multilabels by our best network is shown in Fig. 9.

V. DISCUSSION

Automated architecture search methods would enable domain scientists to apply CNN techniques with little or no deep learning experience. In this article, we have introduced an SLGE-based random search with an early stopping strategy to automatically search for efficient CNN architectures for both single-label and multilabel RS scene classification tasks. We experimented with the proposed method on RGB aerial image datasets, NWPU-RESISC45 and AID, and on multispectral satellite image datasets, EuroSAT and BigEarthNet. Through comparing and analyzing the experimental results with the state-of-the-art transfer learning (feature extraction and fine-tuning) models and the baseline network, we evaluated the effectiveness of the proposed SLGE-based random search with an early stopping strategy for RS scene classification. The results showed that the proposed automated architecture search method discovered architectures, which could offer a promising performance in classifying RS image scenes, particularly multispectral satellite image scenes.

As shown in Table VI, in the four-band RGB-NIR space of the EuroSAT dataset, our best discovered network outperformed the baseline network by approximately 1.0% classification accuracy rate, whereas, in the RGB space of EuroSAT, our best discovered network achieved a competitive performance that was only 0.2% lower than the baseline. It is clear that, since the baseline was pretrained on RGB images (ImageNet), this might be the reason why it showed a 0.2% accuracy rate over our best discovered network in the RGB space of EuroSAT but trailed behind our best discovered network in the RGB-NIR space by 1.0%. Moreover, in the RGB-NIR space of the BigEarthNet multilabel dataset, our best discovered network gained a performance boost in F_1 score, recall, and precision metrics with respect to the state-of-the-art models reported in [65], which is relatively higher compared to fine-tuning the baseline. Table VII shows that, in the RGB-NIR space of BigEarthNet, our best discovered network significantly improved the performance of the state-of-the-art models on BigEarthNet by approximately 12.1% in F_1 score, 9.2% in recall, and 8.5% in precision compared to the baseline, which showed approximately 8.2%, 5.7%, and 4.0% in F_1 score, recall, and precision, respectively. Specifically, in Table VII, our best discovered network outperformed the baseline by F_1 score of 3.9%, recall of 3.5%, and precision of 4.5% in the RGB-NIR space of BigEarthNet. Even in case using four-band RGB-NIR space of the multispectral image dataset, the proposed SLGE-based method achieved relatively better performance in the RS image classification task.

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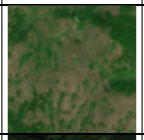
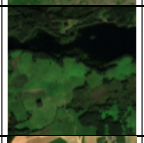
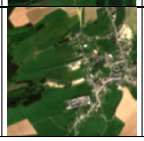
Image	True Multi-Label	Predicted Multi-Label	Image	True Multi-Label	Predicted Multi-Label
	Arable land; Inland waters; Coniferous forest; Mixed forest	Arable land; Inland waters; Coniferous forest; Mixed forest; Land principally occupied by agriculture		Natural grassland and sparsely vegetated areas; Broad-leaved forest; Transitional woodland-shrub	Natural grassland and sparsely vegetated areas; Broad-leaved forest; Transitional woodland-shrub
	Arable land; Urban fabric; Complex cultivation patterns; Land principally occupied by agriculture; Pastures	Arable land; Urban fabric; Complex cultivation patterns; Land principally occupied by agriculture; Pastures		Coniferous forest; Broad-leaved forest; Inland waters; Land principally occupied by agriculture; Mixed forest; Pastures	Broad-leaved forest; Land principally occupied by agriculture; Mixed forest; Pastures; Inland waters; Transitional woodland-shrub
	Agro-forestry areas; Pastures; Transitional woodland-shrub; Arable land;	Agro-forestry areas; Pastures; Transitional woodland-shrub		Complex cultivation patterns; Arable land; Mixed forest; Pastures; Urban fabric	Complex cultivation patterns; Arable land; Mixed forest; Pastures; Urban fabric

Fig. 9. Example of the BigEarthNet RGB images with the true multilabels and the predicted multilabels by our best network. The green labels are true labels correctly predicted; the blue labels are true labels that were not predicted; and the red labels are wrongly predicted labels.

The performance comparisons on RGB aerial image datasets (NWPU-RESISC45 and AID) indicate that the proposed SLGE-based method competed with the state-of-the-art transfer learning models and the baseline network. For example, in Table III, with about 91% less memory footprint, our best discovered network achieved 0.2% improvement of classification accuracy over the best feature extraction method, DLGFF [47], and approximately 1.3% improvement over the best fine-tuned model, Fusion of VGGNet-16 & CaffeNet [74]. Also, in Table IV, the best discovered network improved the performance of the fine-tuned models by approximately 0.8% classification accuracy rate and competed with the state-of-the-art feature extraction methods. However, it can be seen from the confusion matrices (see Figs. 5 and 7) that the proposed method demonstrated a poor performance on some scene classes, particularly, *palace* and *church* classes in NWPU-RESISC45 (see Fig. 6) and *bareland*, *desert*, *square*, and *centre* classes in AID (see Fig. 8). The reason is that most of the examples of these classes exhibit very high interclass similarities in textural and structural characteristics. Searching for CNNs that can solve this problem is one of the future research topics.

VI. Conclusion

In this article, we presented an NAS-based method for the classification of RS image scenes. Using SLGE-based architecture representation, we explored random search with early stopping strategy as an architecture search technique to automatically search for CNN architectures for RS scene classification task. The proposed method could generate multipath Inception-like cells with skip-connections similar to human-expert designs as building blocks for CNN architectures. The experimental results on four RS scene classification datasets, two RGB aerial image benchmarks (NWPU-RESISC45 and AID), and two multispectral satellite image benchmarks (EuroSAT and BigEarthNet) suggest that the autogenerated CNN models demonstrated promising performance in classifying multispectral satellite image scenes. With fewer parameters, the best discovered networks significantly improved the state-of-the-art models on EuroSAT single-label and BigEarthNet multilabel classification tasks. We are hoping that the architecture learning strategy proposed here will suggest something inspiring and opens a new research path for any type of neural search architecture strategies, including reinforcement learning and evolutionary algorithm, to search for CNN architectures for other RS image classification tasks, such as semantic segmentation and object detection.

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References

[1] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[2] S. S. Durbha, K. R. Kurte, and U. Bhangale, "Semantics and high performance computing driven approaches for enhanced exploitation of Earth observation (EO) data: State of the art," *Proc. Nat. Acad. Sci., India A, Phys. Sci.*, vol. 87, no. 4, pp. 519–539, Dec. 2017.

[3] S. F. Smillie, "Automatic target recognition: Some considerations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-2, no. 2, pp. 187–191, Mar. 1966.

[4] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[5] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[6] J. Song, S. Gao, Y. Zhu, and C. Ma, "A survey of remote sensing image classification based on CNNs," *Big Earth Data*, vol. 3, no. 3, pp. 232–254, Jul. 2019.

[7] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[8] Y. Sun, B. Xue, M. Zhang, and G. G. Yen, "Completely automated CNN architecture design based on blocks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 4, pp. 1242–1254, Apr. 2019.

[9] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*. Red Hook, NY, USA: Curran Associates, 2012, pp. 1097–1105.

[10] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. 22nd ACM Int. Conf. Multimedia*, Nov. 2014, pp. 675–678.

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[11] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun, "OverFeat: Integrated recognition, localization and detection using convolutional networks," 2013, *arXiv:1312.6229*. [Online]. Available: http://arxiv.org/abs/1312.6229

[12] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–14.

[13] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 4278–4284.

[14] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[16] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[17] J. A. Richards and X. Jia, "Sources and characteristics of remote sensing image data," in *Remote Sensing Digital Image Analysis: An Introduction*. Berlin, Germany: Springer, 1999, pp. 1–38.

[18] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 3320–3328.

[19] A. Canziani, A. Paszke, and E. Culurciello, "An analysis of deep neural network models for practical applications," 2016, *arXiv:1605.07678*. [Online]. Available: http://arxiv.org/abs/1605.07678

[20] N. M. Noor, A. Abdullah, and M. Hashim, "Remote sensing UAV/drones and its applications for urban areas: A review," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 169, no. 1, p. 12003, 2018.

[21] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," *J. Mach. Learn. Res.*, vol. 20, no. 55, pp. 1–21, 2019.

[22] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8697–8710.

[23] E. Real, A. Aggarwal, Y. Huang, and Q. V. Le, "Regularized evolution for image classifier architecture search," in *Proc. AAAI*, 2019, pp. 4780–4789.

[24] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–13.

[25] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Sep. 2019.

[26] C. Broni-Bediako, Y. Murata, L. H. B. Mormille, and M. Atsumi, "Evolutionary NAS with gene expression programming of cellular encoding," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2020, pp. 2670–2676.

[27] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *J. Mach. Learn. Res.*, vol. 13, pp. 281–305, Feb. 2012.

[28] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[29] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019.

[30] G. Sumbul, M. Charfuelan, B. Demir, and V. Markl, "BigEarthNet: A large-scale benchmark archive for remote sensing image understanding," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 5901–5904.

[31] F. Hutter, L. Kotthoff, and J. Vanschoren, Eds., *Automated Machine Learning—Methods, Systems, Challenges*. Cham, Switzerland: Springer, 2019.

[32] H. Cai, T. Chen, W. Zhang, Y. Yu, and J. Wang, "Efficient architecture search by network transformation," in *Proc. 32nd AAAI Conf. Artif. Intell.*, 2018, pp. 2787–2794.

[33] Z. Zhong, J. Yan, W. Wu, J. Shao, and C.-L. Liu, "Practical block-wise neural network architecture generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2423–2432.

[34] S. Xie, H. Zheng, C. Liu, and L. Lin, "SNAS: Stochastic neural architecture search," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–17.

[35] C. Liu *et al.*, "Progressive neural architecture search," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 19–34.

[36] L.-C. Chen *et al.*, "Searching for efficient multi-scale architectures for dense image prediction," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 8699–8710.

[37] S. Xie, A. Kirillov, R. Girshick, and K. He, "Exploring randomly wired neural networks for image recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1284–1293.

[38] M. Wistuba, A. Rawat, and T. Pedapati, "A survey on neural architecture search," 2019, *arXiv:1905.01392*. [Online]. Available: http://arxiv.org/abs/1905.01392

[39] J. S. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 2546–2554.

[40] D. Golovin, B. Solnik, S. Moitra, G. Kochanski, J. Karro, and D. Sculley, "Google vizier: A service for black-box optimization," in *Proc. 23rd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2017, pp. 1487–1495.

[41] L. Li *et al.*, "A system for massively parallel hyperparameter tuning," in *Proc. Mach. Learn. Syst.*, vol. 2, 2020, pp. 230–246.

[42] L. Li and A. Talwalkar, "Random search and reproducibility for neural architecture search," in *Proc. Conf. Uncertainty Artif. Intell.*, 2019, pp. 367–377.

[43] K. Yu, C. Sciuto, M. Jaggi, C. Musat, and M. Salzmann, "Evaluating the search phase of neural architecture search," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 1–16.

[44] H. Pham, M. Guan, B. Zoph, Q. Le, and J. Dean, "Efficient neural architecture search via parameters sharing," in *Proc. 35th Int. Conf. Mach. Learn.*, vol. 80, Jul. 2018, pp. 4095–4104.

[45] R. Luo, F. Tian, T. Qin, E. Chen, and T.-Y. Liu, "Neural architecture optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 7816–7827.

[46] Y. Guo, J. Ji, X. Lu, H. Huo, T. Fang, and D. Li, "Global-local attention network for aerial scene classification," *IEEE Access*, vol. 7, pp. 67200–67212, May 2019.

[47] Q. Zhu, Y. Zhong, Y. Liu, L. Zhang, and D. Li, "A deep-local-global feature fusion framework for high spatial resolution imagery scene classification," *Remote Sens.*, vol. 10, no. 4, p. 568, 2018.

[48] D. Zeng, S. Chen, B. Chen, and S. Li, "Improving remote sensing scene classification by integrating global-context and local-object features," *Remote Sens.*, vol. 10, no. 5, p. 734, May 2018.

[49] R. Dong, D. Xu, L. Jiao, J. Zhao, and J. An, "A fast deep perception network for remote sensing scene classification," *Remote Sens.*, vol. 12, no. 4, p. 729, Feb. 2020.

[50] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," 2015, *arXiv:1508.00092*. [Online]. Available: http://arxiv.org/abs/1508.00092

[51] K. Nogueira, O. A. B. Penatti, and J. A. dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognit.*, vol. 61, pp. 539–556, Jan. 2017.

[52] S. Jiang, H. Zhao, W. Wu, and Q. Tan, "A novel framework for remote sensing image scene classification," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 3, pp. 657–663, 2018.

[53] A. Zeggada, F. Melgani, and Y. Bazi, "A deep learning approach to UAV image multilabeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 694–698, May 2017.

[54] S. Koda, A. Zeggada, F. Melgani, and R. Nishii, "Spatial and structured SVM for multilabel image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5948–5960, May 2018.

[55] A. Alshehri, Y. Bazi, N. Ammour, H. Almubarak, and N. Alajlan, "Deep attention neural network for multi-label classification in unmanned aerial vehicle imagery," *IEEE Access*, vol. 7, pp. 119873–119880, 2019.

[56] Y. Hua, L. Mou, and X. X. Zhu, "Recurrently exploring class-wise attention in a hybrid convolutional and bidirectional LSTM network for multi-label aerial image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 188–199, Mar. 2019.

[57] Y. Hua, L. Mou, and X. X. Zhu, "Relation network for multilabel aerial image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4558–4572, Jul. 2020.

[58] Y. Liu and C. Huang, "Scene classification via triplet networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 220–237, Jan. 2018.

[59] R. Stivaktakis, G. Tsagkatakis, and P. Tsakalides, "Deep learning for multilabel land cover scene categorization using data augmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1031–1035, Jul. 2019.

[60] G. Sumbul and B. Demir, "A deep multi-attention driven approach for multi-label remote sensing image classification," *IEEE Access*, vol. 8, pp. 95934–95946, 2020.

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[61] J. Chen *et al.*, "Convolution neural network architecture learning for remote sensing scene classification," 2020, *arXiv:2001.09614*. [Online]. Available: http://arxiv.org/abs/2001.09614
[62] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," in *Proc. 5th Int. Conf. Learn. Represent.*, 2017, pp. 1–16.
[63] R. Caruana, S. Lawrence, and L. Giles, "Overfitting in neural nets: Backpropagation, conjugate gradient, and early stopping," in *Proc. 13th Int. Conf. Neural Inf. Process. Syst.*, 2000, pp. 381–387.
[64] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8024–8035.
[65] G. Sumbul *et al.*, "BigEarthNet dataset with a new class-nomenclature for remote sensing image understanding," 2020, *arXiv:2001.06372*. [Online]. Available: http://arxiv.org/abs/2001.06372
[66] J. Howard and S. Gugger, "Fastai: A layered API for deep learning," *Information*, vol. 11, no. 2, p. 108, Feb. 2020.
[67] L. N. Smith, "A disciplined approach to neural network hyper-parameters: Part 1—Learning rate, batch size, momentum, and weight decay," 2018, *arXiv:1803.09820*. [Online]. Available: http://arxiv.org/abs/1803.09820
[68] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2015, pp. 1–15.
[69] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn.*, vol. 97, 2019, pp. 6105–6114.
[70] G. Cheng, Z. Li, X. Yao, L. Guo, and Z. Wei, "Remote sensing image scene classification using bag of convolutional features," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1735–1739, Oct. 2017.
[71] Y. Yu and F. Liu, "Dense connectivity based two-stream deep feature fusion framework for aerial scene classification," *Remote Sens.*, vol. 10, no. 7, p. 1158, Jul. 2018.
[72] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.
[73] J. Li, D. Lin, Y. Wang, G. Xu, and C. Ding, "Deep discriminative representation learning with attention map for scene classification," 2019, *arXiv:1902.07967*. [Online]. Available: http://arxiv.org/abs/1902.07967
[74] Y. Yu and F. Liu, "Aerial scene classification via multilevel fusion based on deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 287–291, Feb. 2018.
[75] Y. Li, Q. Wang, X. Liang, and L. Jiao, "A novel deep feature fusion network for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 5484–5487.
[76] Y. Yu and F. Liu, "A two-stream deep fusion framework for high-resolution aerial scene classification," *Comput. Intell. Neurosci.*, vol. 2018, Jan. 2018, Art. no. 8639367.
[77] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[78] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.
[79] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[80] G. Sumbul and B. Demir, "A novel multi-attention driven system for multi-label remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 5726–5729.



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